

Agent Design Principles — A Five-Level Learning Ladder

Ten principles ranked by load-bearing-ness (how directly each forces a build decision), wired into one connected-agent OODA loop.

L1 DECIDE — The Rational Core			
	Bayesian Epistemology · Bayes, Pearl	BayesianBeliefAgent	Beliefs are probabilities; revise on every observation.
	Expected Utility · Von Neumann, Savage	DecisionTheoryAgent	The scoring rule: pick the action with the best expected payoff.
	Bounded Rationality · Simon	BoundedRationalityAgent	Budgets, early stopping, satisficing — 'good enough', not perfect.
L2 OPERATE — The Loop and the Goal			
	OODA / Cybernetics · Wiener, Boyd	OODAController	The control skeleton: sense - decide - act - observe - repeat.
	Teleology / Means-Ends · Aristotle, STRIPS	TeleologyPlanner	Start from the goal; work backward to candidate actions.
	Belief-Desire-Intention · Bratman	BDIState	Split what it knows, wants, and has committed to.
L3 IMPROVE & REASON — Learning and Thinking			
	Reinforcement Learning · Skinner, Sutton & Barto	ReinforcementLearner	Reward outcomes; credit assignment tunes what the agent values.
	Dual-Process (System 1/2) · Kahneman	DualProcessRouter	Decide when to think harder: fast reaction vs. deliberation.
	Dialectic + Pragmatism · Hegel, Socrates, Peirce	DialecticCritic	Generator-vs-critic: test an idea against its opposite; keep what works.
L4 STRATEGIZE — Strategy Among Others			
	Game Theory · Nash, Schelling	GameTheoryAgent	Best response when other agents act strategically; avoid collisions.
L5 CONSTRAIN — The Values Layer (add-on)			
	Deontology vs. Consequentialism · Kant, Mill	constraint filter	Rules vs. outcomes — guardrails checked before an intention commits.
	Stoic control · Epictetus · Occam's Razor	scoping / prior	Boundary heuristic + simplicity prior that shape, not drive, decisions.

How to read it: climb the ladder. L1 makes one good decision; L2 wraps it in a loop with a goal and memory; L3 lets the agent learn and reason about when to think harder; L4 adds other strategic agents; L5 is the values/guardrail layer that constrains the rest. The runnable network lives in `code_samples/connected_agents.py`.